

2026 Machine-Learning Investment Assignment

Instructor: Prof. Gavin Feng

Course weight 20%

Team Up to three

Release After Class 4

Due 15 October 2026

Submission deadline: 12:00 noon, Hong Kong time (UTC+8).

Overview. Forecast weekly stock excess returns and construct a long-only portfolio using the supplied simulated data. Compare OLS, LASSO, and either random forest or boosting in Q1–Q5.

Data and timing

The panel covers 260 simulated weeks for 80 stock labels from the July 2023 Hang Seng Index universe. Dates and names are labels; numerical observations are simulated, with stronger predictive signals than in real stock returns.

Period	Weeks	Stock-week rows	Next-week returns
Training	1–78	6,240	Visible
Validation	79–104	2,080	Visible
Private evaluation	105–260	12,480	Hidden

Use "project_data_handout_2026.csv" or "project_data_handout_2026.xlsx" (identical 20,800 rows). The dictionary lists 17 dimensionless predictors: six market and eleven stock/sector signals. For example, `momentum_short` is simulated, not calculated from `ret_next_week`.

Predictors and `rf_weekly` are available at the forecast date. The next week's total return, `ret_next_week`, is already aligned with that row; do not shift it. Predict

$$y = \text{ret_next_week} - \text{rf_weekly}.$$

Use only the 17 predictors as inputs, excluding identifiers, dates, split labels, and outcomes.

Q1. Data and prediction benchmark (10 points)

- **Sample counts and target statistics.** Report stocks, weeks, and rows by split, plus the training target's mean and sample standard deviation.
- **Figure 1: Predictor correlations.** Plot the training correlation matrix for all 17 predictors. Identify one strongly correlated pair and report its correlation.
- **Time split and benchmark.** Explain why random stock-week splits are unsuitable. Define the zero-excess-return forecast and calculate its validation MSE.

Hint. The zero-excess-return benchmark forecasts a total return of `rf_weekly`; it differs from Q5's equal-weight portfolio. Pool the same 2,080 validation rows for every model:

$$R_{\text{OOS}}^2 = 1 - \frac{\sum (y - \hat{y})^2}{\sum y^2}.$$

Q2. OLS and LASSO

(20 points)

Fit preprocessing and pooled OLS/LASSO on Weeks 1–78 with all 17 predictors and an intercept. Standardize with training means and standard deviations. Apply the same scaler to Weeks 79–104 and select settings there. Keep weeks intact in any additional time-based validation.

- **Table 1: Prediction results.** Report training MSE, validation MSE, and validation OOS R^2 for the zero forecast, OLS, and LASSO. Add Q3's model later.
- **Figure 2: LASSO penalty selection.** Plot validation MSE over a small penalty grid. Report the selected penalty and number of nonzero coefficients.
- **Table 2: Coefficients.** Compare standardized OLS and selected LASSO coefficients for all 17 predictors, with a separate intercept row. Which signals does LASSO retain, and does selection improve validation accuracy?

Hint. Five to ten penalties suffice; choose the penalty with the lowest validation MSE. Larger penalties usually retain fewer signals. No estimation derivation is required.

Q3. One nonlinear model

(15 points)

Fit **either random forest or boosting** with the same predictors and split. Explain when it could improve on a linear model.

- **Model settings and selection.** Compare settings by validation MSE. Report those tried and the selected specification.
- **Table 1: Prediction results.** Add the selected model and explain whether it improves validation prediction.

Hint. Try four to six settings. For random forest, vary depth or minimum leaf size at a fixed tree count. For boosting, vary tree count at a fixed learning rate and shallow depth. Set a random seed. Extra models are optional; complexity alone earns no credit.

Q4. Characteristic importance

(15 points)

Apply Class 5's drop-to-zero method to the selected nonlinear model. Keep it fixed, set one standardized validation predictor to zero, and recalculate OOS R^2 . Repeat for all 17 predictors.

- **Figure 3: Characteristic importance.** Importance is original OOS R^2 minus OOS R^2 after replacement. Plot the five largest values; retain negative values.
- **Figure 4: Response profile.** Vary the most important standardized predictor from its training 10th to 90th percentile. Hold other inputs at their validation values and average predictions.
- **Comparison with LASSO.** Explain one difference between these results and LASSO's selected variables.

Hint. Standardized zero is the training mean. Use the same scaler for fitting and importance. Importance is predictive, not causal; correlated signals can overlap.

Q5. Forecasts and portfolios**(15 points)**

Each validation week, rank all 80 stocks by predicted excess return, breaking ties by ascending `asset_id`. Apply both rules to each model:

- **Long-short:** equally weight the top 16 and short the bottom 16. The spread is the top group's average return minus the bottom group's average return.
- **Long-only:** equally weight the top 16, with zero elsewhere (within the 10% position cap).

Table 3: Validation portfolios. For each model, report the mean weekly long-short spread and the long-only portfolio's annualized Sharpe ratio, information ratio versus equal weight, and maximum drawdown. Add the 80-stock equal-weight benchmark (N/A for its long-short spread). Does the best prediction model produce the best portfolio?

Final portfolio choice and risk. Choose a model and long-only rule from validation evidence: keep the top-16 rule or propose another. Explain your choice and identify or add its row in Table 3. Report its largest validation-week sector weight, naming the sector and week, and discuss one material risk. Refit preprocessing and the model on Weeks 1–104, then keep both fixed. Use each private week's predictors to form weights for Weeks 105–260.

Weight constraints. Weights must be nonnegative, sum to 100%, and not exceed 10% per stock. Rebalance each of the 156 private weeks, starting from equal weights at the end of Week 104. Ignore transaction costs. Long-short results belong in the report, not the weight file.

Portfolio measures

Use 52 weeks per year and sample standard deviations. Weekly returns are r_p for the portfolio, r_f for the risk-free asset, and r_{EW} for the rebalanced 80-stock equal-weight portfolio:

$$\text{Sharpe ratio} = \sqrt{52} \frac{\text{mean}(r_p - r_f)}{\text{sd}(r_p - r_f)},$$

$$\text{Information ratio} = \sqrt{52} \frac{\text{mean}(r_p - r_{EW})}{\text{sd}(r_p - r_{EW})}.$$

For maximum drawdown, start wealth at 1 and compound total returns. Report the largest percentage fall from a previous peak as a nonpositive number. For grading, set ratios with standard deviation below 10^{-12} to zero, including equal weight's information ratio against itself.

Submission requirements

Submit three files with the same lead student's ID, replacing `leadstudentid` (e.g., 12345678):

1. **Report:** "`EF5560_leadstudentid_report.pdf`". At most eight pages including the cover, plus an optional one-page appendix. List all members' names and student numbers on the cover. Use Q1–Q5 headings, page numbers, and the three result tables. Layout is your choice; "`report_outline_2026.txt`" is optional.
2. **Weights:** "`EF5560_leadstudentid.csv`".
Follow "`portfolio_weight_rules_2026.txt`". Keep the template's 156 dates and 80 ticker columns in order. Use decimal weights with at least eight decimal places and run either supplied checker. Equal weight is a valid example, not a required strategy.
3. **Code appendix:** "`EF5560_leadstudentid_code.zip`", at most 10 MB. Include R/Python scripts or notebooks and helper source files used for the report and weights. Exclude data and installed packages. No separate README is required.

Implementation details. End with the main script/notebook name, software/package versions, and random seeds. Code is for verification; length and programming style are not graded.

Format and units. Export from Word, LaTeX, or similar software with selectable, copyable text and tables. Use plots, not code/console screenshots. Label units: decimal returns, squared decimal returns for MSE, and decimal OOS R^2 . Keep at least six significant digits for MSE and four decimal places for other summary statistics; calculate from unrounded values.

AI use statement. AI coding assistance is allowed. End the report with the tools used and one or two checks your team performed, or state that no AI tools were used.

Grading criteria

Component	Points
Q1–Q5: 10 + 20 + 15 + 15 + 15	75
Implementation: software/package versions 2; random seeds 1; valid weight CSV 2	5
Private investment performance: Sharpe 8; information ratio 6; maximum drawdown 6	20
Total	100

Correct analysis and clear interpretation earn credit even when results are negative.

For each private metric, rank valid submissions from worst to best and award equally spaced scores from 50% to 100% of its points. Average tied ranks. If all tie or only one submission is valid, award 75%. Drawdown closer to zero ranks higher. Invalid files receive zero private-performance points.